

C

DHTSystem

```
package dht
import java.util.ArrayList
import java.util.List

void addNode(NodeThread node)
nodes.add(node);
node.start();
void buildDiscoveryChain()
void printChain()
```

C

Main

```
package dht

static void main(String[] args) throws InterruptedException
system.addNode(n1);
system.addNode(n2);
system.addNode(n3);
system.buildDiscoveryChain();
system.printChain();
n1.receiveMessage(putMsg);
Thread.sleep(500);
n3.receiveMessage(getMsg);
```

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Message

```
package dht
Type type
String key
String value
Node origin
```

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Node

```
package dht

void receiveMessage(Message msg);
long getId();
```

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NodeThread

```
package dht
import java.util.*
import java.util.concurrent.*
final long id

NodeThread(long id)
void run()
processMessage(msg);
} catch (InterruptedException e)
Thread.currentThread().interrupt();
void processMessage(Message msg)
switch (msg.type)
localStore.put(msg.key, msg.value);
forwardToPeers(msg);
void forwardToPeers(Message msg)
for (Node peer : peers)
peer.receiveMessage(msg);
void addPeer(Node peer)
void receiveMessage(Message msg)
mailbox.offer(msg);
synchronized (System.out)
```